

Name: _____

Class: _____

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12

HSC ASSESSMENT TASK 3

JUNE 2014

MATHEMATICS Extension 1

Time Allowed: 70 minutes

Instructions:

- Write your name and class at the top of each page.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a **new page**.
- Standard integrals can be found on the last page.

1. What is the derivative of $y = \cos^{-1}\left(\frac{1}{x}\right)$ with respect to x ?

(A) $\frac{-1}{\sqrt{x^2-1}}$

(B) $\frac{-1}{x\sqrt{x^2-1}}$

(C) $\frac{1}{\sqrt{x^2-1}}$

(D) $\frac{1}{x\sqrt{x^2-1}}$

2. The number N of animals in a population at time t years is given by $N=100 + Ae^{kt}$ for constants $A > 0$ and $k > 0$. Which of the following is the correct differential equation?

(A) $\frac{dN}{dt} = k(N-100)$

(B) $\frac{dN}{dt} = -k(N+100)$

(C) $\frac{dN}{dt} = -k(N-100)$

(D) $\frac{dN}{dt} = k(N+100)$

3. If $f(x) = 1 - \cos \frac{x}{2}$ what is the inverse function $f^{-1}(x)$?

(A) $f^{-1}(x) = 2 \cos^{-1}(1-x)$

(B) $f^{-1}(x) = \frac{1}{2} \cos^{-1}(1-x)$

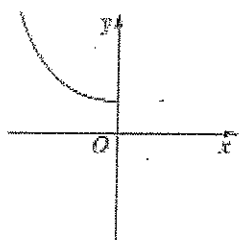
(C) $f^{-1}(x) = \frac{1}{2} \cos^{-1}(1+x)$

(D) $f^{-1}(x) = 2 \cos^{-1}(1+x)$

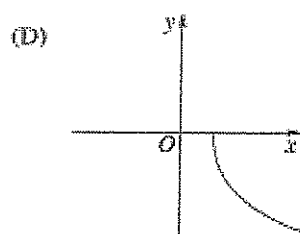
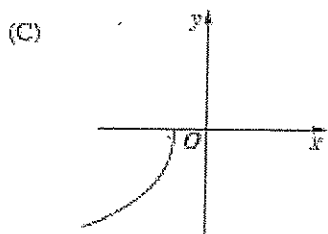
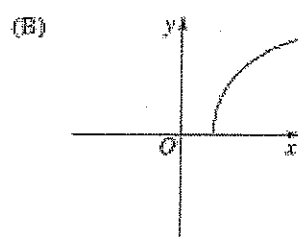
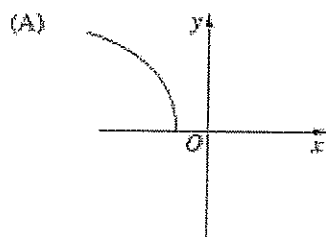
4. What is the domain and range of $y = \cos^{-1}\left(\frac{3x}{2}\right)$?

- (A) Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$. Range: $0 \leq y \leq \pi$
 (B) Domain: $-1 \leq x \leq 1$. Range: $0 \leq y \leq \pi$
 (C) Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$. Range: $-\pi \leq y \leq \pi$
 (D) Domain: $-1 \leq x \leq 1$. Range: $-\pi \leq y \leq \pi$

5. The diagram of the graph $y = f(x)$



Which diagram shows the graph of $y = f^{-1}(x)$?



Question 6 (8 marks)

- a) Write the exact value of :
- i) $\sin^{-1} \frac{\sqrt{3}}{2}$ 1
 - ii) $\sin^{-1}(\sin(-\frac{\pi}{4}))$ 1
- b) Simplify $\cos \left(2 \cos^{-1} \frac{\sqrt{3}}{2} \right)$ 2
- c) Write the equation $\ln x + \ln y^2 = 3$ without logarithms 1
- d) Solve for x : $\log_{10}(x^2) + \log_{10} x = 1$ 1
- e) Find $\frac{d^2}{dx^2} (e^{x^2})$ 2

Start a new page**Question 7 (8 marks)**

- a) Find the derivative of $\sin^{-1} x + \cos^{-1} x$ 1
and hence find the exact value of $\sin^{-1} x + \cos^{-1} x$
(Show all working) 2
- b) Differentiate the following with respect to x :
- i) $g(x) = \ln x^2 - e$ 1
 - ii) $h(x) = \ln \left(\frac{e^x - 1}{e^x + 1} \right)$ 2
(leaving your answer in simplified exact form)
 - iii) $y = \cos^{-1}(-x) + \cos^{-1}(x)$ 2

Start a new page

Question 8 (9 marks)

- a) Sketch the curve $y = \sin^{-1} 3x$. 2
- b) Differentiate $e^{\tan^{-1} x}$ with respect to x 1
- c) i) Find $\frac{d}{dx}(xe^x - e^x)$ 1
- ii) Hence, or otherwise, find $\int_0^1 xe^x dx$ 2
- d) Find the inverse function for $g(x) = \sqrt{5-x} - 1$ and state the domain and range for the inverse 3

Start a new Page

Question 9 (8 marks)

- a) Find the equation of the tangent to the curve $y = 4 \sin^{-1}\left(\frac{x}{2}\right)$ at the point where $x = 1$. (Leave in exact form) 3
- b) Find $\int \frac{\ln 2x}{x} dx$ using the substitution $u = \ln 2x$, or otherwise 2
- c) Find the exact value of $\cos\left(\sin^{-1}\left(\frac{5}{13}\right) + \sin^{-1}\left(\frac{4}{5}\right)\right)$ 3

Start a new page

Question 10 (8 marks)

a) Differentiate $\tan^{-1} e^{2x}$ and hence find $\int_0^{\frac{1}{2}} \frac{4e^{2x}}{1+e^{4x}} dx$ as an exact answer 3

b) The rate at which a body cools in air is proportional to the difference between the temperature, T , of the body and the constant surrounding temperature, S . this can be expressed as $\frac{dT}{dt} = k(T - S)$ where t is time in minutes and k is a constant.

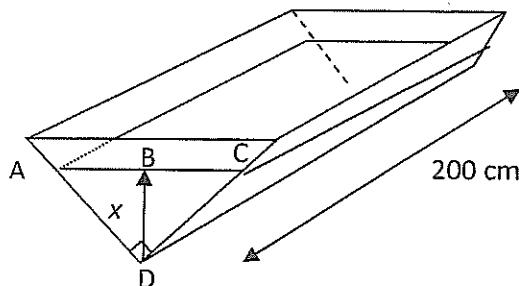
i. Show that $T = S + Be^{kt}$ where B is a constant, is a solution of the above equation 1

ii. If a particular body cools from 100° to 80° in 30 minutes, find the temperature of the body after a further 30 minutes, given the surrounding temperature remains a constant 25° . Give your answer to the nearest degree. 4

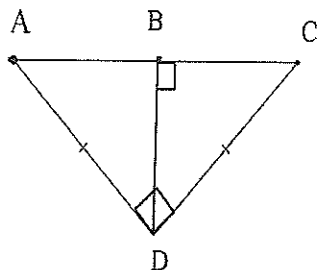
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Question 11 (9 marks)

- a) A water trough is 200 cm long and has the cross section of a right-angled isosceles triangle. B is the midpoint of the line AC. 'x' is the depth of the water in the trough.



(i)



Prove that $AD = DC$ $BD = BC$.

2

- (ii) Show that when the depth of the water is x cm, the volume of the water in the tank is $200x^2 \text{ cm}^3$, explaining all steps. 1
- (iii) Water is poured in at a constant rate of 5 litres per minute. Find the rate at which the water level is rising when the depth is 30 cm (1 litre = 1000 cm^3) 2

b) Differentiate $\left(\tan^{-1}\left(\frac{x}{3}\right)\right)^2$, and hence find the exact value of $\int_0^{\sqrt{3}} \frac{\tan^{-1}\left(\frac{x}{3}\right)}{x^2 + 9} dx$ 2

c) By writing $y = \tan^{-1}\sqrt{x}$ in the form $x = f(y)$, show that $\frac{dy}{dx} = \frac{1}{2\sqrt{x}(1+x)}$ 2

D A A A D

Q6 (a) $\sin^{-1} \frac{\sqrt{3}}{2} = \pi/3$

(ii) $\sin^{-1}(\sin^{-\pi/4}) = -\pi/4$

(b) $\cos(2 \cos^{-1} \sqrt{3}/2)$



$\cos^{-1} \sqrt{3}/2 = \alpha$
 $\cos \alpha = \sqrt{3}/2$

$\cos 2\alpha = 2 \cos^2 \alpha - 1$
 $= 2 (\sqrt{3}/2)^2 - 1$
 $= 3/2 - 1$
 $= 1/2$

(c) $\ln x + \ln y^2 = 3$
 $\ln(xy^2) = 3$
 $e^3 = xy^2$

(d) $\log_{10}(x^3) + \log_{10} x = 1$
 $\log_{10} x^3 = \log_{10} 10$
 $x^3 = 10$
 $x = 2.154$

(e) $y = e^{x^2}$
 $y = e^u \quad u = x^2$
 $\frac{dy}{du} = e^u \quad \frac{du}{dx} = 2x$
 $\frac{dy}{dx} = 2xe^{x^2}$

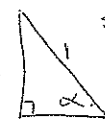
$\frac{d^2y}{dx^2} = 2x(2xe^{x^2}) + 2e^{x^2}$
 $= e^{x^2}(4x^2 + 2)$
 $= 2e^{x^2}(2x^2 + 1)$

Q7 (a) $f(x) = \sin^{-1} x + \cos^{-1} x$
 $f'(x) = 0$
 $\therefore$ gradient constant

$0 + \pi/2 = C$
 $\sin^{-1} x + \cos^{-1} x = \pi/2$

OR

$\sin^{-1} x = \alpha, \cos^{-1} x = \pi/2 - \alpha$
 $\sin \alpha = x \quad \cos(\pi/2 - \alpha)$



$\sin^{-1} x + \cos^{-1} x = \alpha + \pi/2 - \alpha$
 $= \pi/2$

(b) (i) $g(x) = \ln x^2 - e$
 $g'(x) = 2/x$

(ii) $h(x) = \ln \left(\frac{e^x - 1}{e^x + 1} \right)$

$h(x) = \ln u$
 $= \frac{1}{u}$
 $= \frac{e^x + 1}{e^x - 1} \cdot \frac{2e^x}{(e^x + 1)^2}$
 $= \frac{2e^x}{(e^x - 1)(e^x + 1)}$

$u = (e^x - 1)(e^x + 1)^{-1}$
 $u' = (e^x - 1)(-1)(e^x + 1)^{-2} (e^x)$
 $+ e^x (e^x + 1)^{-1}$

$= \frac{-e^x(e^x - 1)}{(e^x + 1)^2} + \frac{e^x}{e^x + 1}$
 $= \frac{-e^x(e^x - 1) + e^x(e^x + 1)}{(e^x + 1)^2}$

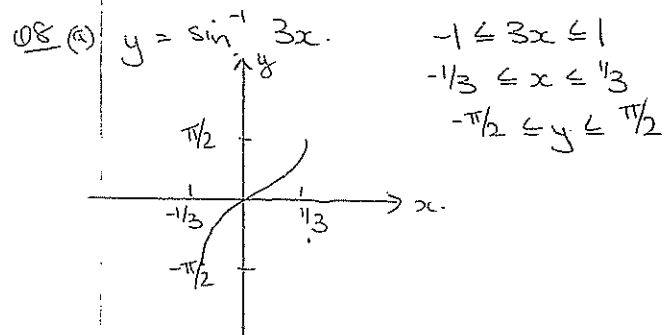
$= \frac{2e^x}{(e^x + 1)^2}$

OR $h(x) = \ln(e^x - 1) - \ln(e^x + 1)$

$= \frac{e^x}{e^x - 1} - \frac{e^x}{e^x + 1}$

$= \frac{e^x(e^x + 1) - e^x(e^x - 1)}{(e^x - 1)(e^x + 1)} = \frac{2e^x}{(e^x - 1)(e^x + 1)}$

Q7 (b)(iii) $y = \cos^{-1}(-x) + \cos^{-1}(x)$
 $y' = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}}$
 $= 0.$



(b) $y = e^{\tan^{-1} x}$
 $\frac{dy}{dx} = e^u \cdot \frac{du}{dx}$
 $= e^u \cdot \frac{1}{1+x^2}$
 $\frac{dy}{dx} = \frac{e^{\tan^{-1} x}}{1+x^2}$

(c) (i) $\frac{d}{dx} (xe^x - e^x) = e^x + xe^x - e^x$
 $= xe^x$

(ii) $\int_0^1 xe^x dx = [xe^x - e^x]_0^1$
 $= (e - e) - (0 - 1)$
 $= 1$

(d)

Q8 (d) $y = \sqrt{5-x} - 1$

$x = \sqrt{5-y} - 1$
 $x+1 = \sqrt{5-y}$
 $(x+1)^2 = 5-y$
 $(x+1)^2 - 5 = -y$
 $y = 5 - (x+1)^2$
 $= 5 - x^2 - 2x - 1$
 $= -x^2 - 2x + 4$

D: $x \leq 5$

R: $y \geq -1$



D: $x \geq -1$ R: $y \leq 5$

Q9 (a) $y = 4 \sin^{-1} \left(\frac{x}{2} \right)$
 $y' = \frac{4}{\sqrt{2^2 - x^2}}$
 $= \frac{4}{\sqrt{4 - x^2}}$

at $x=1$
 $m = \frac{4/\sqrt{3}}{3}$

at $x=1$ $y = 4 \sin^{-1} (1/2)$
 $= 2\pi/3$

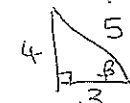
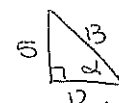
$y - 2\pi/3 = \frac{4\sqrt{3}}{3} (x-1)$
 $3y - 2\pi = 4\sqrt{3} (x-1)$

$4\sqrt{3}x - 4\sqrt{3} + 2\pi - 3y = 0$
 $4\sqrt{3}x - 3y + 2(\pi - 2\sqrt{3}) = 0$

(b) $\int \frac{\ln 2x}{x} dx$
 $= \int u du$
 $= \frac{u^2}{2} + C$
 $= \frac{(\ln 2x)^2}{2} + C$

$u = \ln 2x$
 $du = \frac{1}{x} dx$

(c) $\cos(\sin^{-1}(5/13) + \sin^{-1}(4/5))$ let $\alpha = \sin^{-1} 5/13$ let $\beta = \sin^{-1} 4/5$
 $= \cos(\alpha + \beta)$
 $= \cos \alpha \cos \beta - \sin \alpha \sin \beta$
 $= \frac{12}{13} \cdot \frac{3}{5} - \frac{5}{13} \cdot \frac{4}{5}$



Q10 (a) $\tan^{-1} e^{2x}$
 $\frac{dy}{dx} = \frac{2e^{2x}}{1+(e^{2x})^2}$
 $= \frac{2e^{2x}}{1+e^{4x}}$

$y = \tan u$
 $\frac{dy}{du} = \frac{1}{1+u^2}$

$u = e^{2x}$
 $\frac{du}{dx} = 2e^{2x}$

(ii) $\int_0^{1/2} \frac{4e^{2x}}{1+e^{4x}} dx = 2 [\tan^{-1} e^{2x}]_0^{1/2}$
 $= 2 [\tan^{-1} e - \tan^{-1} 1]$
 $= 2 (\tan^{-1} e - \pi/4)$
 $= 2 \tan^{-1} e - \pi/2$

(b) (i) $\frac{dT}{dt} = k(T-S)$

(a) $T = S + Be^{kt}$
 $\frac{dT}{dt} = k(Be^{kt})$
 $= k(T-S)$

(ii) $T_i = 100$ $T_{30} = 80$ $t = 30 \text{ mins}$ $S = 25$

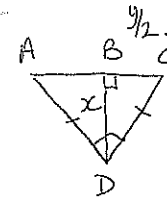
$100 = 25 + Be^0$
 $B = 75$

$80 = 25 + 75e^{k30}$
 $11/15 = e^{k30}$
 $k = -0.0103$
 or $\frac{\ln 11/15}{30}$

$T_{60} =$ $t = 60$ $S = 25$

$T = 25 + 75e^{(-0.0103 \times 60)}$
 $= 65.33^\circ = 65^\circ$

Q11 (i)



ΔACD is right angled isosceles

$\therefore \angle DAC = \angle DCA =$
 $\angle DAC = \angle DCA = 45^\circ$ (angle sum of triangle equals (180°))
 $\angle ADC = 90^\circ$ (given)

(ii) $\tan 45^\circ = \frac{x}{y/2}$

$1 = \frac{2x}{y/2}$

$\frac{y}{2} = x$

$y = 2x$

$\frac{dV}{dt} = 5l$
 $= 5000 \text{ cm}^3$

$V = \frac{1}{2}bh \times 200$
 $= \frac{1}{2}(2x)(x)(200)$

$V = 200x^2$

$\frac{dV}{dt} = 200 \cdot 2x \frac{dx}{dt}$
 at $x = 30$

$5000 = (200)(2)(30) \frac{dx}{dt}$

$\frac{dx}{dt} = 5/12 \text{ cm}^3/\text{min}$

$$(b) i) y = \left(\tan^{-1} \left(\frac{x}{3} \right) \right)^2$$

$$f'(x) = 2 \tan^{-1} \left(\frac{x}{3} \right) \frac{3}{9+x^2}$$

$$= \frac{6 \tan^{-1} \left(\frac{x}{3} \right)}{9+x^2}$$

$$(ii) \int_0^{\sqrt{3}} \frac{\tan^{-1} \left(\frac{x}{3} \right)}{x^2+9} dx = \frac{1}{6} \int \frac{6 \tan^{-1} \left(\frac{x}{3} \right)}{x^2+9} dx$$

$$= \frac{1}{6} \left[\left(\tan^{-1} \frac{x}{3} \right)^2 \right]_0^{\sqrt{3}}$$

$$= \frac{1}{6} \left[\left(\tan^{-1} \frac{\sqrt{3}}{3} \right)^2 - \left(\tan^{-1} 0 \right)^2 \right]$$